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$$\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y^2}{x^4 + y^4}$$
$$\left\{ \begin{array}{l} x = 0 \\ \lim_{(0,y) \rightarrow (0,0)} \frac{0^2 y^2}{0^4 + y^4} = 0 \end{array} \right.$$
$$\left\{ \begin{array}{l} y = x \\ \lim_{(x,x) \rightarrow (0,0)} \frac{x^2 x^2}{x^4 + x^4} = \frac{x^4}{2x^4} = \frac{1}{2} \end{array} \right.$$

$\Rightarrow$ De limiet bestaat niet, want $0 \neq \frac{1}{2}$.

References